Example of Inverter Grade Thyristor Ratings

- V / I rating: 4500V / 3000A.
- Max. Frequency: 20KHz.
- Switching time: 20 to 100μsec.
- On state resistance: 0.5mΩ.

Example of Triac Ratings

- Used in heat / light control, ac motor control circuit
- V / I rating: 1200V / 300A.
- Max. Frequency: 400Hz.
- Switching time: 200 to 400μsec.
- On state resistance: 3.6mΩ.

Example of Power Transistor Ratings

- PT ratings go up to 1200V / 400A.
- PT normally operated as a switch in CE config.
- Max. Frequency: 400Hz.
- Switching time: 200 to 400μsec.
- On state resistance: 3.6mΩ.

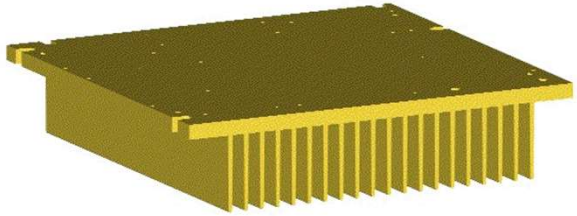
Example of Power MOSFET Ratings

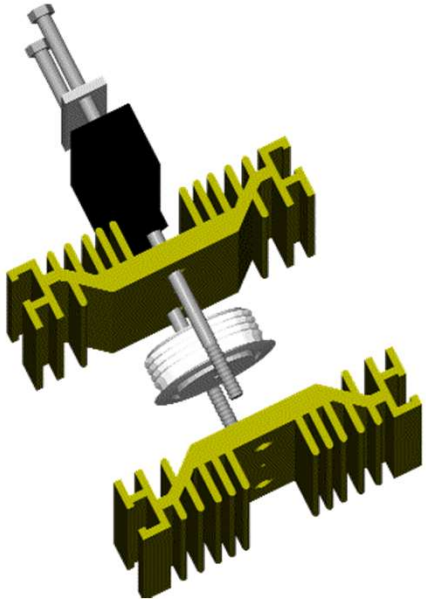
- Used in high speed power converters like inverters & choppers.
- Ratings up to 1000V / 100A.
- Example: MOSFET 800V / 7.5A rating.
- Max. Frequency: 100KHz.
- Switching time: 1.6μsec.
- On state resistance: 1.2mΩ.

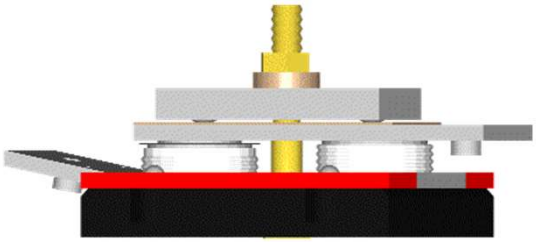
Example of IGBT Ratings

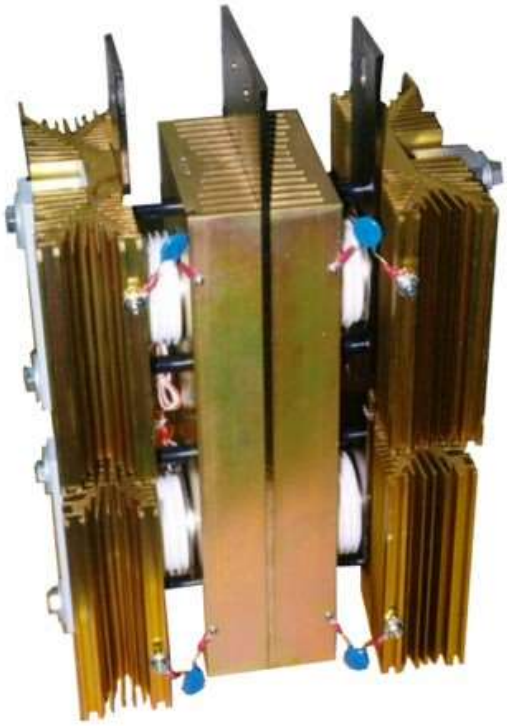
- Used in high voltage / current & high frequency switching power applications (Inverters, SMPS).
- Example: IGBT 2500V / 2400A.
- Max. Frequency: 20KHz.
- Switching time: 5 to μsec.
- On state resistance: 2.3mΩ.

Designing – Heat Sinking

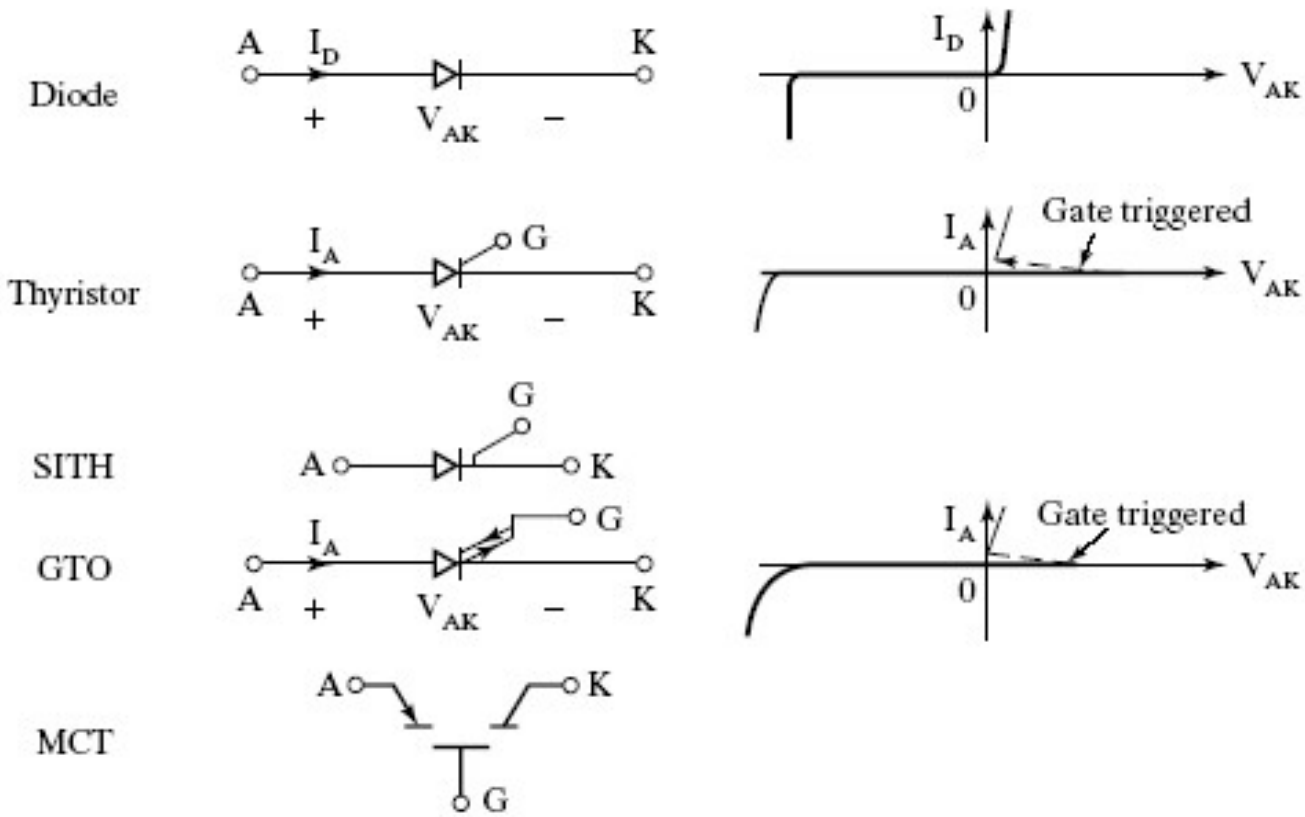




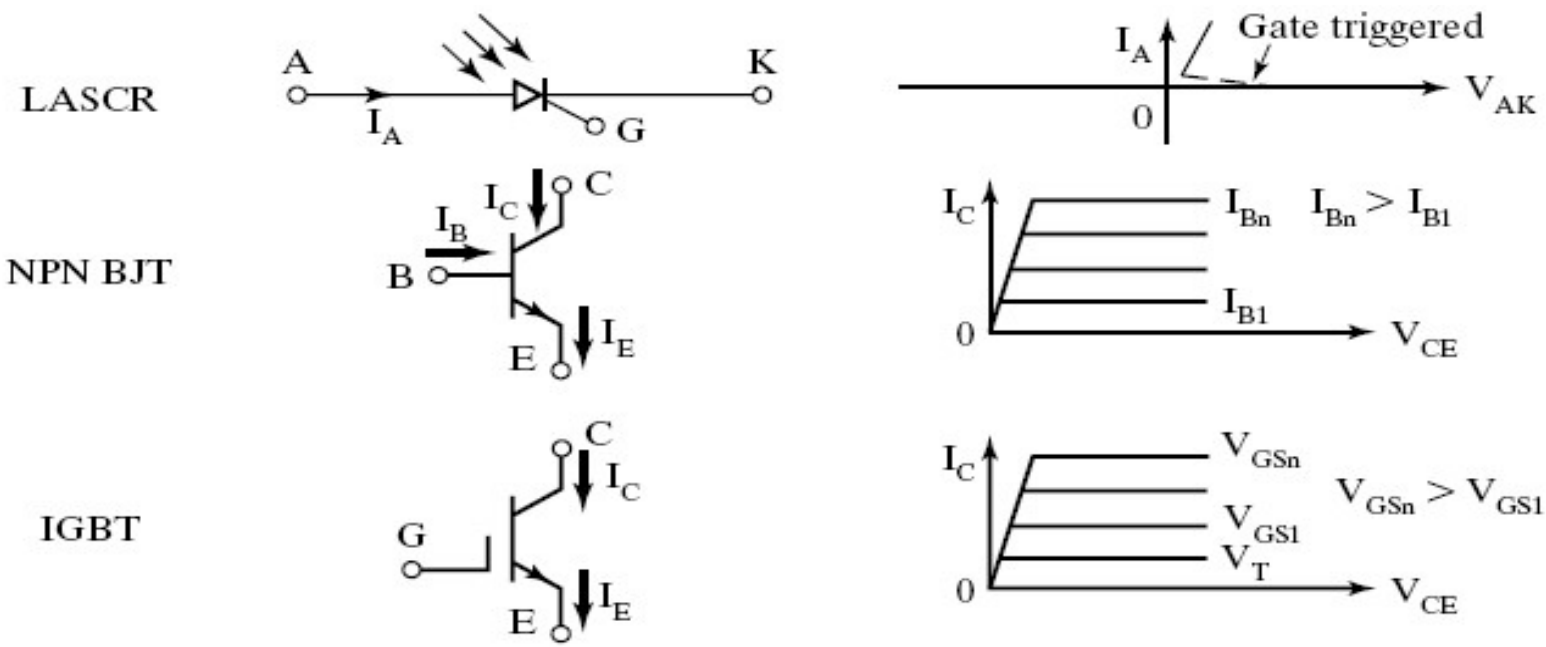




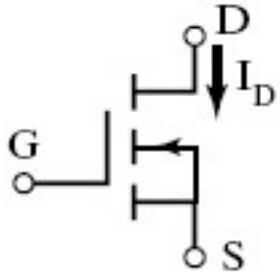
Power Semiconductor Devices, their Symbols & Characteristics

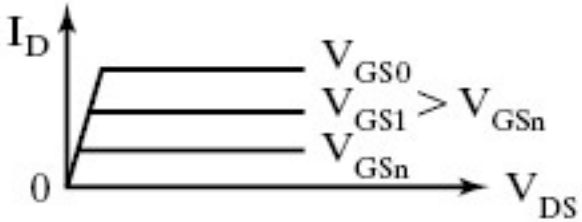


DEVICE SYMBOLS & CHARACTERISTICS



N-Channel
MOSFET





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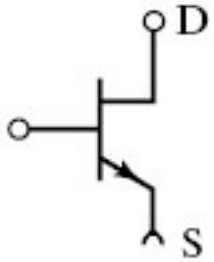
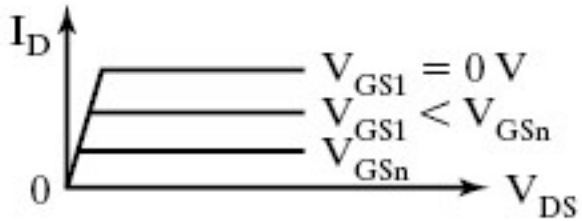
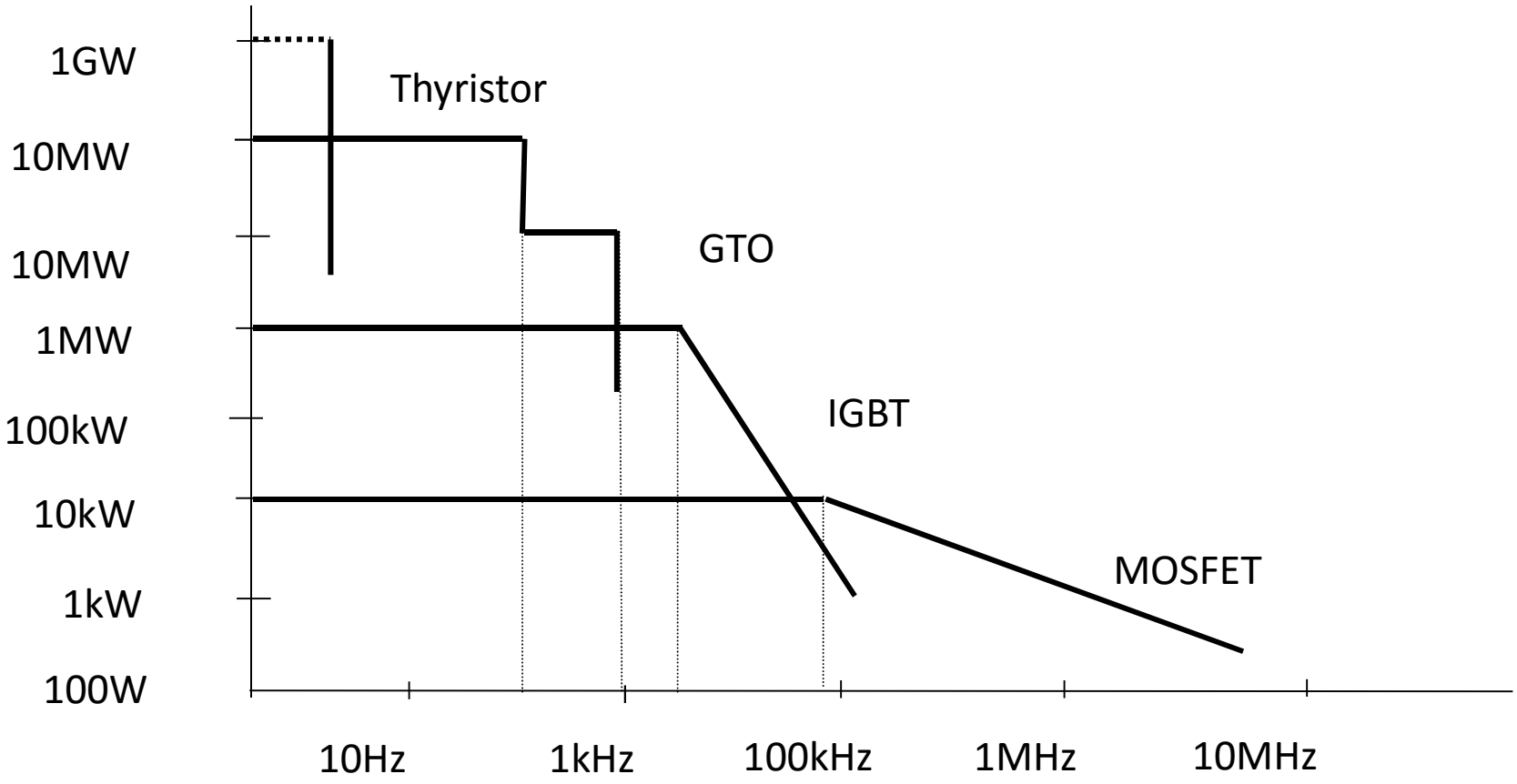




Power Switches: Power Ratings



	Thy	BJT	FET	GTO	IGBT	IGCT
Avail-abilty	Early 60s	Late 70s	Early 80s	Mid 80s	Late 80s	Mid 90's
State of Tech.	Mature	Mature	Mature/ improve	Mature	Rapid improve	Rapid improvem ent
Voltage ratings	5kV	1kV	500V	5kV	3.3kV	6.5kV
Current ratings	4kA	400A	200A	5kA	1.2kA	4kA
Switch Freq.	na	5kHz	1MHz	2kHz	100kHz	1kHz
On-state Voltage	2V	1-2V	I* Rds (on)	2-3V	2-3V	3V
Drive Circuit	Simple	Difficult	Very simple	Very difficult	Very simple	Simple
Comm-ents	Cannot turn off using gate signals	Phasing out in new product	Good performan ce in high freq.	King in very high power	Best overall performanc e.	Replacing GTO